EE446
Lab 2
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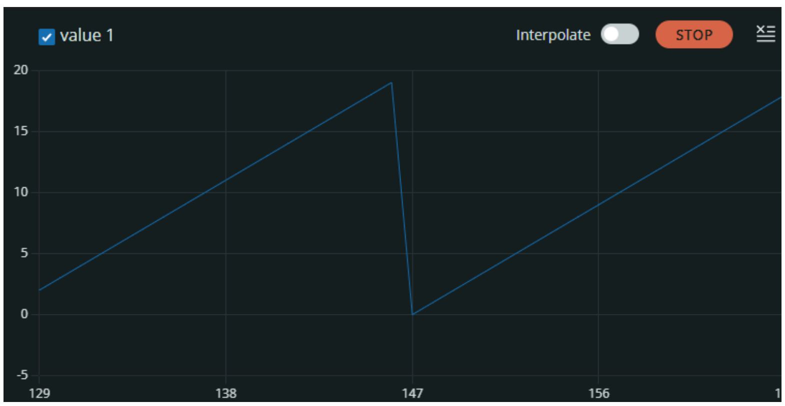
Repository Link: https://github.com/drgusimyth/EE446-Lab2

Task 4:

```
21:33:38.353 -> Serial test started
21:33:39.076 -> Count: 0
21:33:40.083 -> Count: 1
21:33:41.089 -> Count: 2
21:33:42.065 -> Count: 3
21:33:43.042 -> Count: 4
21:33:44.042 -> Count: 5
21:33:45.075 -> Count: 6
21:33:46.042 -> Count: 7
21:33:47.042 -> Count: 8
21:33:48.066 -> Count: 9
21:33:49.060 -> Count: 10
21:33:50.065 -> Count: 11
21:33:51.057 -> Count: 12
21:33:52.061 -> Count: 13
```

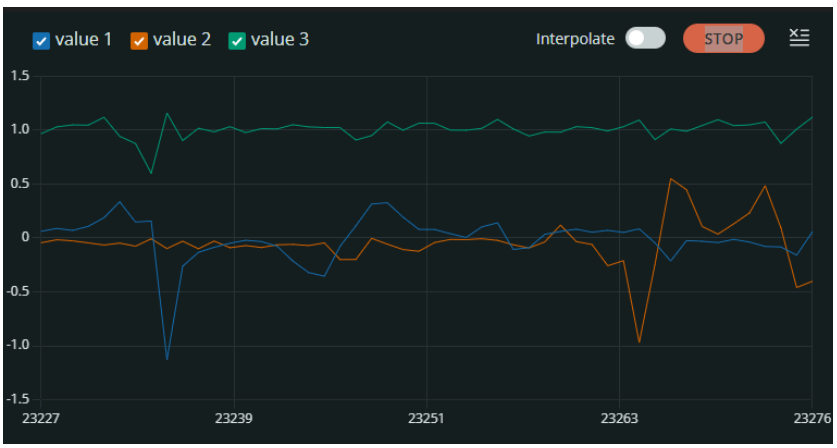


Task 5:
Relatively quiet environment:

When speaking near the board:

When I shouted up close to the board, the greatest change plotted signal was observed.

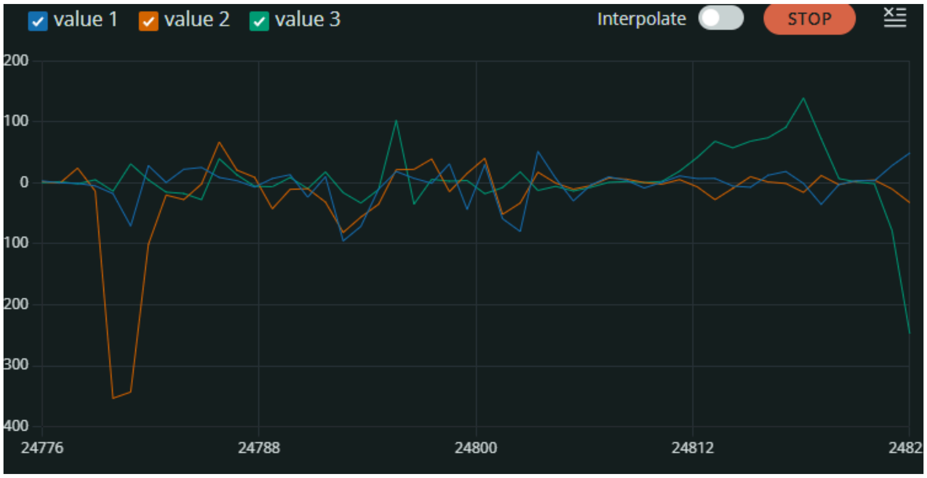
Task 6A:




```
22:30:48.913 -> 0.013,-0.137,1.007
22:30:48.992 -> 0.136,-0.104,1.012
22:30:49.089 -> 0.020,0.004,1.014
22:30:49.227 -> 0.019,-0.003,1.012
22:30:49.321 -> 0.018,0.013,1.013
22:30:49.410 -> 0.021,-0.003,1.013
22:30:49.538 -> 0.027,-0.002,1.012
22:30:49.624 -> 0.018,-0.002,1.013
22:30:49.755 -> 0.020,-0.007,1.014
22:30:49.839 -> 0.025,-0.009,1.012
```

Suddenly moving the board vertically up produced the clearest change in the z acceleration.

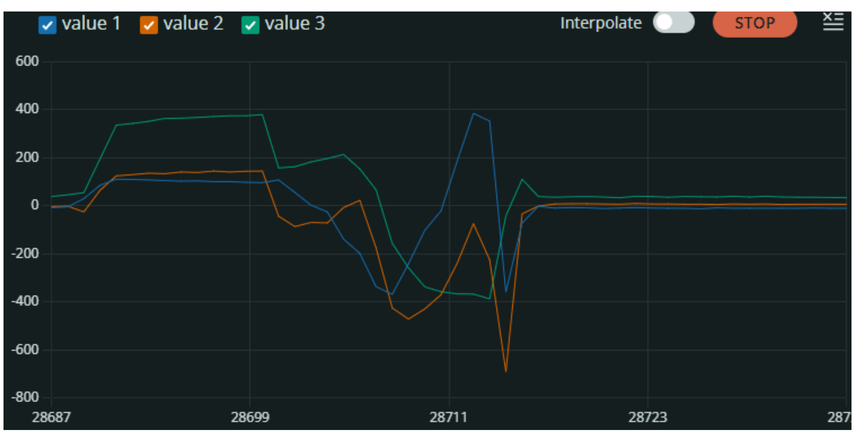
Task 6B:




```
22:58:05.806 -> -0.916,-0.244,-0.488
22:58:05.889 -> -0.671,-0.183,-0.305
22:58:05.996 -> -0.305,-0.610,-0.061
22:58:06.134 -> 0.732,-1.892,0.854
22:58:06.228 -> -0.427,-0.244,-0.183
22:58:06.318 -> -0.488,-0.366,-0.549
22:58:06.444 -> -0.305,-0.183,-0.183
22:58:06.555 -> 0.061,-0.122,0.000
22:58:06.680 -> 0.000,-0.793,0.122
22:58:06.774 -> -0.305,-0.305,-0.305
```

Rotating the board along the x, y, and z axis produced the same magnitude of change in their corresponding signals, with a constant speed of rotation corresponding to a constant value in the signal.

Task 6c:



```
23:03:59.690 -> -11.000,6.000,38.000
23:04:00.000 -> -11.000,8.000,33.000
23:04:00.138 -> -11.000,7.000,37.000
23:04:00.326 -> -11.000,7.000,36.000
23:04:00.506 -> -10.000,6.000,35.000
23:04:00.721 -> -11.000,8.000,39.000
23:04:00.936 -> -12.000,7.000,32.000
23:04:01.119 -> -11.000,8.000,39.000
23:04:01.342 -> -10.000,6.000,37.000
23:04:01.526 -> -11.000,8.000,38.000
```

Putting my phone next to the board resulted in a significant change in all three signals.

Task 7:
Stable temperature and humidity:

```
23:53:17.988 -> Temperature (C): 27.17 | Humidity (%): 42.84
23:53:19.999 -> Temperature (C): 27.18 | Humidity (%): 42.85
23:53:21.040 -> Temperature (C): 27.17 | Humidity (%): 42.84
23:53:22.152 -> Temperature (C): 27.17 | Humidity (%): 42.85
23:53:23.186 -> Temperature (C): 27.17 | Humidity (%): 42.85
23:53:24.259 -> Temperature (C): 27.17 | Humidity (%): 42.89
```

Temperature and humidity after a breathing close to the board:

```
23:54:06.106 -> Temperature (C): 27.12 | Humidity (%): 43.53
23:54:07.209 -> Temperature (C): 27.12 | Humidity (%): 44.09
23:54:08.282 -> Temperature (C): 27.15 | Humidity (%): 45.57
23:54:09.367 -> Temperature (C): 29.02 | Humidity (%): 74.18
23:54:10.425 -> Temperature (C): 28.20 | Humidity (%): 80.40
23:54:11.470 -> Temperature (C): 27.84 | Humidity (%): 80.08
23:54:12.542 -> Temperature (C): 27.77 | Humidity (%): 80.45
23:54:13.616 -> Temperature (C): 28.87 | Humidity (%): 85.88
```

Breathing close to the board resulted in the greatest change in humidity

Task 8:
Stable temperature and pressure values:

```
23:57:09.620 -> Pressure (kPa): 101.394 | Temperature (C): 27.89
23:57:10.655 -> Pressure (kPa): 101.391 | Temperature (C): 27.89
23:57:11.659 -> Pressure (kPa): 101.391 | Temperature (C): 27.89
23:57:12.692 -> Pressure (kPa): 101.392 | Temperature (C): 27.91
23:57:13.697 -> Pressure (kPa): 101.394 | Temperature (C): 27.92
23:57:14.713 -> Pressure (kPa): 101.394 | Temperature (C): 27.91
23:57:15.710 -> Pressure (kPa): 101.396 | Temperature (C): 27.94
23:57:16.755 -> Pressure (kPa): 101.393 | Temperature (C): 27.93
23:57:17.770 -> Pressure (kPa): 101.392 | Temperature (C): 27.94
23:57:18.764 -> Pressure (kPa): 101.396 | Temperature (C): 27.95
```

```
23:57:58.452 -> Pressure (kPa): 101.392 | Temperature (C): 28.93
23:57:59.471 -> Pressure (kPa): 101.392 | Temperature (C): 29.30
23:58:00.489 -> Pressure (kPa): 101.336 | Temperature (C): 29.59
23:58:01.528 -> Pressure (kPa): 101.391 | Temperature (C): 29.01
23:58:02.529 -> Pressure (kPa): 101.397 | Temperature (C): 28.97
23:58:03.582 -> Pressure (kPa): 101.402 | Temperature (C): 28.84
23:58:04.587 -> Pressure (kPa): 101.404 | Temperature (C): 28.83
23:58:05.607 -> Pressure (kPa): 101.402 | Temperature (C): 28.81
23:58:06.605 -> Pressure (kPa): 101.403 | Temperature (C): 28.80
23:58:07.654 -> Pressure (kPa): 101.403 | Temperature (C): 28.78
```

Dropping the board to the ground increased the pressure slightly

Task 9A:

Far from hand:

```
03:22:06.723 -> Proximity: 232
03:22:06.951 -> Proximity: 234
03:22:07.172 -> Proximity: 232
03:22:07.332 -> Proximity: 231
03:22:07.552 -> Proximity: 232
03:22:07.757 -> Proximity: 233
03:22:07.944 -> Proximity: 233
03:22:08.170 -> Proximity: 230
03:22:08.375 -> Proximity: 232
03:22:08.593 -> Proximity: 233
```

Close to hand:

```
03:22:50.851 -> Proximity: 0
03:22:51.037 -> Proximity: 0
03:22:51.218 -> Proximity: 0
03:22:51.438 -> Proximity: 0
03:22:51.669 -> Proximity: 0
03:22:51.828 -> Proximity: 0
03:22:52.056 -> Proximity: 0
03:22:52.274 -> Proximity: 0
03:22:52.462 -> Proximity: 0
03:22:52.639 -> Proximity: 0
```

The value measured the distance from the sensor to the object in front of it

Task 9B:

```
Message (Enter to send message to 'Ardui
03:37:57.900 -> RIGHT
03:38:03.030 -> LEFT
03:38:06.058 -> LEFT
03:38:08.040 -> DOWN
03:38:10.520 -> UP
03:38:12.782 -> RIGHT
03:39:36.604 -> DOWN
03:39:44.455 -> LEFT
03:39:46.480 -> RIGHT
03:39:49.794 -> UP
```

All gestures are rather unreliable and require multiple tries to be detected.

Task 9C:

Baseline lighting:

```
03:48:36.011 -> 17,12,9,36
03:48:36.412 -> 17,12,9,37
03:48:36.784 -> 17,12,9,36
03:48:37.188 -> 17,12,9,36
03:48:37.592 -> 17,12,9,36
03:48:38.053 -> 17,12,9,36
03:48:38.425 -> 17,12,9,36
03:48:38.843 -> 17,12,9,36
03:48:39.257 -> 17,12,9,36
03:48:39.614 -> 17,12,9,35
```

Lighting after moved under lamp, which produced a significant difference:

```
03:49:12.787 -> 16,11,9,35
03:49:13.202 -> 16,11,9,34
03:49:13.578 -> 16,11,9,35
03:49:14.026 -> 16,11,9,34
03:49:14.386 -> 17,12,9,36
03:49:14.825 -> 46,33,41,166
03:49:15.196 -> 3294,3120,2998,3635
03:49:15.625 -> 2312,2070,1995,2856
03:49:16.040 -> 2866,2592,2419,3408
03:49:16.444 -> 2998,2796,2657,3411
```

Task 10:

```
20:22:40.217 -> raw,mic=20,clear=37,motion=0.06,prox=188
20:22:40.217 -> flags,sound=0,dark=0,moving=0,near=0
20:22:40.217 -> state,QUIET_BRIGHT_STEADY_FAR
20:22:40.296 -> raw,mic=139,clear=37,motion=0.06,prox=188
20:22:40.343 -> flags,sound=1,dark=0,moving=0,near=0
20:22:40.343 -> state,NOISY_BRIGHT_STEADY_FAR
20:23:12.725 -> raw,mic=253,clear=203,motion=1.08,prox=0
20:23:12.725 -> flags,sound=1,dark=0,moving=1,near=1
20:23:12.725 -> state,NOISY_BRIGHT_MOVING_NEAR
```

```
20:32:38.021 -> raw,mic=17,clear=0,motion=0.02,prox=0
20:32:38.021 -> flags,sound=0,dark=1,moving=0,near=1
20:32:38.021 -> state,QUIET_DARK_STEADY_NEAR
```

I selected the threshold by first setting them to zero and observing the range of values each modality could take, then I tweaked the threshold so that it corresponded to what is subjectively loud, dark, moving, and near. For motion, I took the norm of the acceleration vector as the measurement quantity.

Task 11:

Magnetic disturbance event:

```
22:50:24.005 -> raw,rh=41.46,temp=27.72,mag=38.72,r=9,g=5,b=5,clear=32
22:50:24.005 -> flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
22:50:24.005 -> event,BASELINE_NORMAL
22:50:24.157 -> raw,rh=41.49,temp=27.70,mag=65.25,r=9,g=5,b=5,clear=32
22:50:24.157 -> flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=0
22:50:24.157 -> event,MAGNETIC_DISTURBANCE_EVENT
```

Light or color change event:

```
22:50:24.532 -> raw,rh=41.49,temp=27.72,mag=81.24,r=2,g=2,b=1,clear=4
22:50:24.532 -> flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
22:50:24.532 -> event,BASELINE_NORMAL
22:50:24.731 -> raw,rh=41.49,temp=27.72,mag=87.42,r=3049,g=2723,b=2574,clear=4097
22:50:24.731 -> flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=1
22:50:24.731 -> event,LIGHT_OR_COLOR_CHANGE_EVENT
```

Magnetic disturbance event:

```
23:00:31.430 -> raw,rh=49.34,temp=28.98,mag=62.38,r=31,g=17,b=12,clear=77
23:00:31.430 -> flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=0
23:00:31.430 -> event,MAGNETIC_DISTURBANCE_EVENT
23:00:31.990 -> raw,rh=65.46,temp=30.05,mag=62.37,r=6,g=3,b=2,clear=13
23:00:32.026 -> flags,humid_jump=1,temp_rise=1,mag_shift=0,light_or_color_change=0
23:00:32.026 -> event,BREATH_OR_WARM_AIR_EVENT
```

(Baseline normal shown in each screenshot)

The thresholds were chosen to be a significant difference between consecutive measurements when an event occurs. The specific values were chosen by observing how much each measurement changed for the corresponding sensors in the previous tasks. The cooldown was programmed as an array of 3 elements corresponding to how much cooldown each event other than baseline has. Every time an event is detected, the element would be set to 4 (2s), and each iteration of the loop would decrease the cooldown by 1, and it is only allowed to fire if its corresponding cooldown is zero.